POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Programming in C (New)

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are key phases of software development life cycle? How do they contribute to successful development of software? 7

   b) What are the properties of good algorithm? Write an algorithm and draw flowchart to find the factorial of number. 8

2. a) Define operator and operand. Explain Arithmetic, Relational and Logical operator with suitable examples. 8

   b) Write a program to read the radius of the circular plot and calculate its area and print 7

   "Park" if area >3000

   "Playground" if $1000 \leq$ area ≤ 3000

   "Garden" if the area is <1000

3. a) Differentiate between while and do-while loop with suitable program. 8

   b) Write a program to input m*n order matrix and check is a given number is present in the matrix or not. If present, display the position of the number in array. 7

OR

What do you mean by array of string? Write a program to check whether a given word is palindrome or not.

4. a) Write a program to read N numbers in an array and find the maximum and minimum number among them also find the average of entered number. 8

   b) Define recursive function. Write a program to find factorial of a given number using recursive function. 7

5. a) Does function return single or multiple value? When and how a function will return single or multiple value? Illustrate with examples. 8

b) What are different DMA functions in C Programming. Explain any two of them with suitable example. 7

OR

Write a program to sort the array using dynamic memory allocation.

6. a) Create structure named Employee. Employee has attributes name, address, post and salary. Input the records for 'n' employees and display information of the employees whose salary is above 50000. 8

b) Explain need of file handling in C Programming. Explain different file opening modes. 7

7. Write short notes on: **(Any two)** 2×5

a) Rules for naming identifiers

b) String handling functions

c) Pointers and Arrays